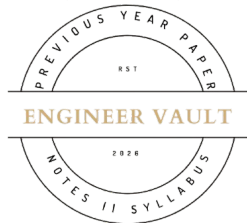


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SUBJECT CODE NO: RNP-8019
FACULTY OF SCIENCE AND TECHNOLOGY
S.E (CSE/CS/CE/AI&DS/AIML/IT/AI) (NEP) Sem - III
Examination May/June-2026
Discrete Mathematics

[Time: 1:30 Hours]**[Max. Marks: 30]**

N. B

Please check whether you have got the right question paper.

- 1) Part A: Question No. 1 is compulsory.
- 2) Attempt any four questions from remaining Part B: Q.2 to Q.7.

PART-A**Q1 Solve any 5 following questions****10**

- a) Define power set.
- b) What is conditional probability?
- c) Define logical equivalence.
- d) What is recursive definition?
- e) Define partial order relation.
- f) State Lagrange's theorem.
- g) What is parity bit?

PART-B**Solve any four questions from the following.**

- | | |
|--|----------|
| Q. 2 Prove De Morgan's Law. | 5 |
| Q. 3 If $P(A) = 0.5, P(B) = 0.3, P(A \cap B) = 0.2$, find $P(A B)P(A B)P(A B)$. | 5 |
| Q. 4 check whether $(p \vee q) \rightarrow r$ is logically equivalent to $(p \rightarrow r) \wedge (q \rightarrow r)$. | 5 |
| Q. 5 Translate: "All students are intelligent" using quantifiers. | 5 |
| Q. 6 Define one-one and onto function with example. | 5 |
| Q. 7 Explain parity check with example. | 5 |

